

Myoclonic seizures subside in the fourth decade in juvenile myoclonic epilepsy

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ABSTRACT

Objective: Our aim was to assess the long-term follow-up of juvenile myoclonic epilepsy (JME), with an emphasis on the course of the myoclonic seizures.

Methods: We enrolled 48 patients with JME (29 F, 19 M; aged 39.9 ± 9.5 years) followed up for a mean of 19.6 ± 5.7 years. The remission for 5 years and relapses were evaluated for all seizure types and the changes in severity/frequency of myoclonia were systematically questioned. The clinical and EEG features, antiepileptic drug (AED) treatment regimen, and systemic and psychiatric comorbid diseases were evaluated.

Results: We found a benign course in 66.6% whereas 16.7% had pseudo-resistance due to problems in treatment or lifestyle. The true-resistant course observed in the remaining 16.7% was significantly associated with psychiatric disorders and the presence of thyroid diseases. In 54.2% of the patients, myoclonia were in remission for a mean duration of 8.4 ± 7.7 years, after an average age of 32.9 ± 9.6 . Of these patients, 6 were on a lower dose of AED in comparison to the dosage needed to control the seizures in the beginning, and 5 patients had stopped AED treatment. None of the latter 11 patients except one relapsed during the follow-up. Furthermore, 21 other patients (43.8%) described substantial alleviation after age 31.3 ± 8.4 in the severity of myoclonia.

Conclusion: Although a great majority of the patients with juvenile myoclonic epilepsy had continuing seizures after a follow-up of 20 years, almost all had either 5-year remission or a substantial alleviation of the myoclonic seizures. *Neurology*® 2008;70:2123-2129

GLOSSARY

AED = antiepileptic drug; **AS** = absence seizures; **CBZ** = carbamazepine; **GTCS** = generalized tonic-clonic seizures; **IGE** = idiopathic generalized epilepsies; **JME** = juvenile myoclonic epilepsy; **LTG** = lamotrigine; **PB** = phenobarbital; **PHT** = phenytoin; **TLE** = temporal lobe epilepsy; **VPA** = valproate.

Juvenile myoclonic epilepsy (JME) is a common syndrome classified among the idiopathic generalized epilepsies (IGE).¹ It has several seizure patterns, with the major characteristic being myoclonic seizures.^{2,3} Generalized tonic-clonic seizures (GTCS) often occur, and one-third of individuals also have absence seizures (AS),⁴ but myoclonia may remain the only seizure type in some individuals. Onset is usually in adolescence but seizures may begin or be diagnosed in the early 20s. Although myoclonic seizures are not specific for JME, their appearance as the cardinal symptom mostly in the morning in a neurologically normal patient makes the diagnosis easy.⁵ The typical abnormality on EEG is bilateral spike or multispikes and wave complexes, at a rate of four to six per second, usually with anterior predominance. Photosensitivity occurs in about 30%, especially in women, but this feature appears to be separately inherited.

Although JME is widely recognized and presumed to be a lifelong trait with a high recurrence rate after withdrawal of antiepileptic drugs (AED), prospective long-term studies documenting this concept are extremely rare. A recent long-term follow-up study of JME included 257 patients initially enrolled for genetic studies.⁶ This multicentric

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study has the main advantage of a large sample size, but they did not focus on the long-term prognosis of the seizure types separately and the follow-up periods as well as the involved clinicians were not uniform.

Our aim was to report the long-term follow-up of patients with JME, with an emphasis on the course of the myoclonic seizures.

METHODS In the initial step of this study, we had retrospectively analyzed the files of 156 patients diagnosed with JME according to the criteria of International League Against Epilepsy⁷ between 1972 and 1996 in our epilepsy outpatient clinic. This epilepsy center is a part of a university hospital and a well-established tertiary referral center for patients with epilepsy in Turkey. On the other hand, due to its convenient location in the center of the metropolis Istanbul, admittance of the patients for primary neurology and emergency departments are also common, constituting at least one half of our cases. The patients having a minimum follow-up duration of 7 years in our institution were included without any other exclusion criteria and the first retrospective results of this survey were published in a national journal at that time.⁸ Afterwards, we continued to follow up these subjects periodically after obtaining their consent. The prospective evaluation began in 1997 and the analyses included all data retrospectively and prospectively collected between 1972 and 2006. If the patients delayed the scheduled visits for more than 1 year, they were invited to a subsequent visit with telephone calls or letters. No specific treatment regimen, laboratory investigations, or EEG controls were planned for them and they were managed according to clinical needs, like all other patients with JME. Lifestyle (sleep patterns, alcohol use) and its changes were closely monitored during the follow-up. Our protocol included counseling about these factors and about their seizure threshold lowering effects. None of these patients reported using alcohol.

All but two patients (both with familial evidence of JGE) had at least one diagnostic EEG and all available EEGs were reviewed by the experienced investigators. Our routine EEG protocol established after 1990 included hyperventilation and intermittent photic stimulation as described in detail previously.⁹ All patients were independently evaluated by at least two of the experienced epileptologists during the long-term follow-up. If the neurologists noted the existence of psychiatric problems, they were further seen by the psychiatry team to clarify the psychiatric diagnosis according to the Diagnostic and Statistical Manual of Mental Disorders-IV.

We initially had enrolled 51 patients, but 2 patients with progressive course who were diagnosed with Lafora disease with skin biopsy and with the identification of EPM2 mutations and one last patient having partial epilepsy with secondary bilateral synchrony on video-EEG were excluded from follow-up. The remaining 48 patients were re-evaluated in 2006.

Remission was defined as ever being seizure-free for an individual seizure type (either myoclonic or GTCS or both) at least for 5 years in any time during the course with or

without AED treatment. Relapse was defined as recurrence of seizures after being seizure-free for at least 5 years.

The history of myoclonia was taken in full detail to outline its severity, besides its monthly frequency. We asked if the myoclonus caused the patient to fall or to drop objects, if it was visible to others or if it was only perceived by the patient. The patient's self-assessment was used basically to rate the myoclonus severity: 0: no myoclonus; 1: milder; 2: same severity compared to that in the first years of JME. In the final analysis, myoclonic seizures were rated as milder if patients' self-assessment was equal to 1 and monthly frequency was 50% less as compared to the frequency in the first 2 years.

The patients were labeled as having benign course (group I) if they had no GTCS and less than two myoclonic seizures monthly. Resistance to AED therapy (group II) was defined as having one or more GTCS or clonic-tonic-clonic seizures per year or more than one myoclonus monthly despite treatment with valproate (VPA) or other appropriate AED for this syndrome with therapeutic drug levels for the last 5 years. We considered lamotrigine, levetiracetam, topiramate, clonazepam phenobarbital (last two drugs were mostly used in the past) and their combinations with VPA as appropriate choices in JME. The patients having seizures due to problems in lifestyle or inappropriate use of seizure aggravating drugs or being noncompliant were grouped as pseudo-resistant (group III). We also evaluated fluctuations during the course from benign to resistant and vice versa.

Other evaluated features were seizure types, ages at onset of all seizure types, personal and family history, results of physical and neurologic examinations, findings of all EEGs, imaging studies when available, psychiatric involvement, comorbid diseases, and treatment and response to the AED with known aggravating effects such as carbamazepine (CBZ), phenytoin (PHT), and other confounding factors of seizure recurrence during long-term follow-up. The treatment regimen used in the last 5 years was grouped as follows: 1) no treatment, 2) VPA monotherapy, 3) other monotherapy (such as phenobarbital [PB], lamotrigine [LTG], topiramate, levetiracetam, clonazepam), 4) reasonable polytherapy, and 5) inappropriate therapy (with CBZ and PHT).

We used Fisher exact tests to compare categorical variables and Student *t* test to compare continuous variables. A Kaplan-Meier curve evaluating the cumulative probability of achieving myoclonus freedom in relation to course was generated using SPSS software.

RESULTS We enrolled 48 patients (29 female, 19 male) with a mean current age of 39.9 ± 9.5 years who had been followed up for a mean of 19.6 ± 5.7 years (median: 18.8, range 10–37 years) with a mean visit count of 21.6 ± 14.3 excluding telephone calls. We noted a tendency to delay the visits in 73% of the patients. The table presents clinical, EEG, and other relevant findings of these patients with JME.

During follow-up, 31 patients showed 5-year remission for myoclonic seizures or GTCS (13 for both seizure types). Remission for AS only but not for the other two seizure types was not taken into account.

Table Clinical and EEG features of patients with JME

Characteristic	Value
Sex, F/M, %	60.4/39.6
Age at onset, y, mean $\pm$ SD, median	14.4 $\pm$ 2.9, 14
Myo (n = 48), mean $\pm$ SD, median	14 $\pm$ 2.9, 14
AS (n = 12), mean $\pm$ SD, median	14.4 $\pm$ 1.3, 14
GTCS (n = 46), mean $\pm$ SD, median	15.7 $\pm$ 4.3, 15.5
Duration of epilepsy, y, mean $\pm$ SD, median	26.2 $\pm$ 8.8, 26.4
Febrile seizure (%)	7 (14.6)
Systemic and neurologic disorders (%)	11 (22.9)
Thyroid disease*	5
SLE	1
Ischemic cardiac diseases, DM type II	3 (both in 1pt)
Migraine with aura	1
Temporal lobe epilepsy	1
Deceased,† n	2
Psychiatric disorders* (%)	15 (31.3)
Obsessive-compulsive disorder	1
Histrionic personality disorder	1
Depressive disorders	10
Psychotic disorders	2*
Generalized anxiety disorder	1
Family history of epilepsy (%)	19 (39.6)
Focal findings (clinical or EEG evidence), n	8
Photosensitivity (clinical or EEG evidence), n	11
Normalization of EEG on medication after the initial EEG with diagnostic GSW, n	27
Aggravation of the GSW in EEG during hyperventilation, n	16
Abnormal neuroimaging finding	5
Cerebellar metastatic lesion*	1*
Cerebellar atrophy	2
Cerebral atrophy	1
Ventricular enlargement	1

*Two patients out of 32 with benign course (group I) versus 3 patients out of 8 in resistant group (group II) have thyroid disease (Fisher exact test $p = 0.046$).

†Two patients with JME were deceased during the follow-up. One of them died due to an invasive thyroid cancer with cerebellar metastasis at the age of 72 and the other was morbidly obese and had well-controlled JME with atypical psychosis which was also under control for many years. He was found dead in his sleep with unknown etiology at age 51 years.

*Seven patients out of 32 with benign course (group I) vs 6 patients out of 8 in resistant group (group II) have psychiatric disorders (Fisher exact test $p = 0.009$). The remaining two patients with psychiatric disorders belong to the pseudo-resistant group III.

JME = juvenile myoclonic epilepsy; Myo = myoclonic seizures; AS = absence seizures; GTCS = generalized tonic-clonic seizures; SLE = systemic lupus erythematosus; DM = diabetes mellitus; GSW = generalized spike-waves.

Sixteen of these 31 patients showed relapses for GTCS or myoclonia with or without triggering factors. The remission and relapses for seizure types in relation to the treatment groups are shown in detail in figures 1 and 2. The triggering factors for seizures such as sleep deprivation or dose reduction of AED did not have specific influence on the relapsing seizure types. The number of relapses with GTCS (9 patients) was higher than the number of relapses with myoclonus (4 patients).

Thirty-two patients (20 female; 62.5% vs 12 male; 37.5%) had benign course (group I) whereas 8 patients (6 female; 75% vs 2 male; 25%) had resistant course (group II) at the end of the follow-up. We also noted pseudo-resistance in 8 patients (group III; 16.7%) (3 female; 37.5%) due to an inadequate lifestyle in two patients, low compliance in one patient, and insufficient AED treatment in five patients due to financial problems (two patients on PB monotherapy, one patient on PHT monotherapy, and two patients on low dose VPA).

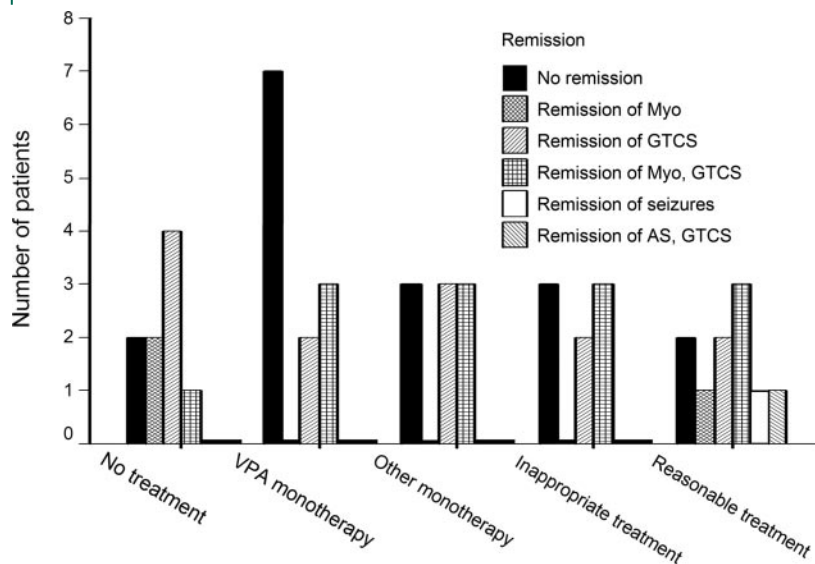
Seven patients (21.9%) of group I with benign course vs 6 patients (75%) of the resistant group II had psychiatric disorders (Fisher exact test $p = 0.009$). The remaining two patients with psychiatric disorders (table) belong to the pseudo-resistant group. Two patients of group I (6.3%) vs 3 patients of resistant group (37.5%) had thyroid disease (Fisher exact test $p = 0.046$).

The gender distribution reported above, delay of diagnosis (mean: 2 years in group I vs mean 3.3 years in group II), and presence of photoparoxysmal response (21.9% in group I vs 12.5% in group II) and other EEG findings were not significantly associated with AED resistance. The presence of all three types of seizures (15.6% in group I vs 50% in group II) was not quite significant for resistant course ($p = 0.059$; Fisher exact test). The difference of the family history of epilepsy (28.1% in group I vs 62.5% in group II) is also not significant, but these results are inconclusive due to the small sample sizes.

In the final analysis, 8 (25%) patients from group I and none from group II discontinued AED treatment, 16 (50%) from group I and only 1 (12.5%) from group II were on a monotherapy regimen, whereas 8 (25%) from group I and 7 (87.5%) patients from group II were using polytherapy.

Twenty-nine patients (60.4%) had received PHT or CBZ in combination with other drugs with a clear aggravation of myoclonic seizures in 44.8%, an improvement in 37.9%, and no change in 17.3%. We did not observe any pa-

Figure 1 Number of patients who ever had a 5-year remission of seizure types in relation to the treatment groups at the end of the study

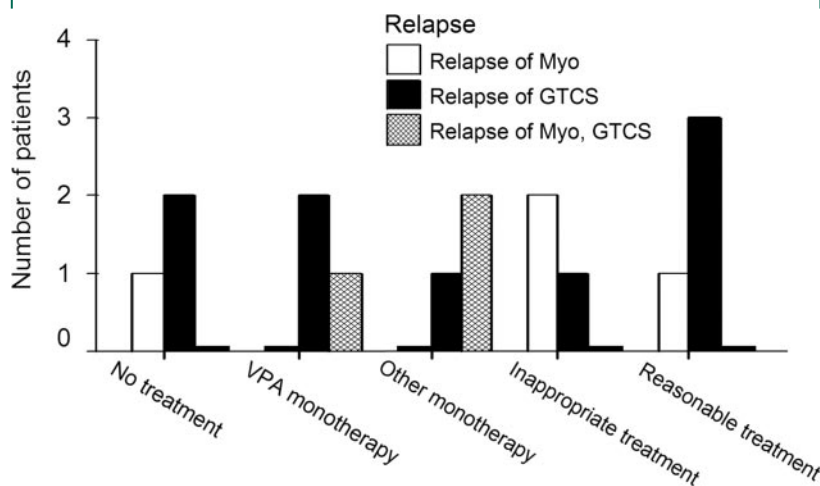


myo = myoclonic seizures; GTCS = generalized tonic-clonic seizures; AS = absence seizures; VPA = valproate.

tient with aggravation of myoclonia under LTG use, but LTG was used in only two patients in our series.

We noted a fluctuating course in three female patients, all having a benign course in the beginning but showing worsening after years. The first one had a new onset thyroid disorder, the second one had DM and congestive heart failure, and the last one had coincidental temporal lobe epilepsy. There were no other reasons to explain these fluctuations. These patients were grouped according to their final situation.

Figure 2 Number of patients who ever had a relapse after having 5-year remission in relation to the treatment groups at the end of the study



myo = myoclonic seizures; GTCS = generalized tonic-clonic seizures; VPA = valproate.

Four patients had remission of AS and another four had continuing AS, which were not causing problems in daily life. The prognosis of AS remained obscure even after this close follow-up in the remaining four of the patients; it could not be clearly decided whether they have had further absence seizures.

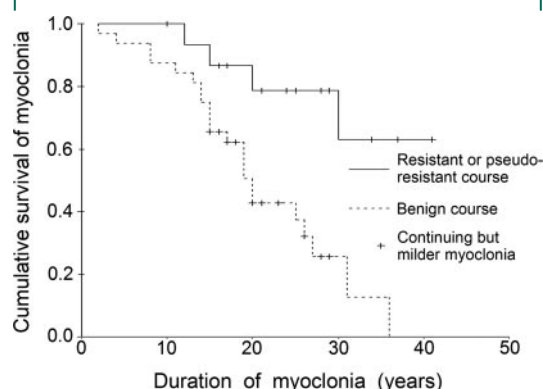
In 26 patients (54.2%; 16 female), myoclonic seizures were in remission in the last follow-up visit (mean duration for remission: 8.4 ± 7.7 with a median of 5.4 years) after an average age of 32.9 ± 9.6 (median: 33) at the start of remission. Of these patients, 15 were using a stable dosage of AED, but 6 were using a lower dose of AED in comparison to the dosage needed to control the seizures in the beginning and the last 5 patients had stopped AED treatment. None of these latter 11 patients except one relapsed, and this single patient who relapsed had GTCS.

The medication regimen adjustments made according to clinical needs during the last 5 years were as follows: 27 patients were using the same dose of the same drug (3 patients were on the same two drugs); 3 subjects were using the same drug or drugs with a lower dose and the treatment was discontinued in 9 subjects (from duotherapy in 2 and from monotherapy in 6). The prescribed drug or drugs was changed in the remaining subjects: from duotherapy to monotherapy in 3 cases, from monotherapy to another monotherapy in one patient, and finally from polytherapy to another polytherapy in 5 cases (all the latter 5 patients belong to the resistant group).

Furthermore, 21 other patients (43.8%; 13 female) described substantial alleviation after age 31.3 ± 8.4 in the severity of myoclonus. Our questioning clearly showed that they had experienced more severe and mostly unprovoked jerks mostly in the morning in the beginning, causing them to drop objects they were holding. The jerks were visible to observers in the beginning; after years the seizures became milder (with unchanged or lower AED dosages or even without any AED) and were hardly seen by observers and occurred mostly at night. Five patients perceived the mild jerks as an inner vibration after years. Falling attacks occurred only in the beginning in 4 cases and in a few of their jerks. Only one patient, aged 41 years, was experiencing severe myoclonic seizures having similar severity and frequency under AED treatment as at the onset of the epilepsy 22 years ago.

Only 4 patients without AED treatment (2 of them were over 40 years) had remission of both myoclonia and GTCS in the final analysis for a

Figure 3 Kaplan-Meier curve showing the cumulative probability of myoclonus freedom according to the course



X axis represents the duration of myoclonic seizures (current age – age at onset of myoclonia). The benign course represents group I described in Methods and resistant or pseudo-resistant course represents both groups II and III for convenience. The censored cases (+) showed patients without myoclonus remission but having milder myoclonus at the last follow-up contact.

mean of 5 years. There were 22 patients aged more than 40 years of age at the final evaluation; 11 of them were totally seizure-free since 5 years. The remaining patients were still experiencing milder myoclonic seizures (6 patients), GTCS and myoclonic seizures (3 patients), GTCS and absence seizures (1 patient), and all three types of seizures (1 patient).

Kaplan-Meier curve evaluating the cumulative probability of achieving myoclonus freedom in relation to the course is shown in figure 3.

DISCUSSION Although there have been many studies dealing with every possible aspect of JME in the last two decades, long-term follow-up studies are rare in the literature. A recent report indicated that JME was perhaps lifelong⁶ in a larger series from various centers over a mean follow-up of 11 ± 6 years. This study is 8 years shorter than our study, which was done in a homogenous group seen in a single epilepsy center.

In two patients with JME with a follow-up duration of 20 years, myoclonic seizures decreased under the same drug regimen after 30 years of age.¹⁰ We showed a substantial alleviation of the myoclonic seizures in the fourth decade for the first time in the literature.

Only a few anecdotal case reports of JME in old age were presented.¹¹⁻¹³ The rarity of patients with JME among the patients with epilepsy aged more than 40 years led us to design this long-term observational study. To have a long-term follow-up duration in the same center, we started the prospective part of this study in 1997, with 48 patients with JME

having at least 7 years of retrospective follow-up in our center.

The striking deterioration observed in some patients with JME taking PHT or CBZ has not been explained.¹⁴ In our series, 60.4% of patients had received PHT or CBZ as add-on therapy at some time during the JME course with an aggravation of myoclonic seizures in 44.8%, an improvement in 37.9%, and no change in 17.3%. It is not easy to change the therapy, which was found useful for the patients on VPA and those who reported an improvement on either CBZ or PHT refused to withdraw the drugs. The usefulness of these drugs in some selected JME cases was also reported by others as add-on treatment, especially for GTCS.¹⁵ Thus, despite recent advances, much remains to be discovered about this common epilepsy syndrome and careful clinical follow-up is needed.

The coexistence of JME with cryptogenic temporal lobe epilepsy (TLE) in one of our cases is interesting to note and this association was previously mentioned in a few case reports only.^{16,17} Our patient had a clear picture of JME with benign course initially. Afterwards TLE appeared with temporal lobe spikes in the EEG. Although these two syndromes retained their inherent electroclinical profile, we could not control her seizures after the emergence of TLE, which did not respond to VPA, but had better seizure control with CBZ.

Three cases of premature death among 170 consecutive JME cases were reported¹⁸ and the authors concluded that serious mental disorders seem to be risk factors for unexpected death. We observed two deaths, one due to metastatic cancer unrelated to epilepsy. The second patient, a very obese 51-year-old man with well-controlled JME and with an atypical psychosis, was found dead in his sleep, although both of his disorders were under control for years.

We found true resistance to AED in 16.7% of this series and our findings are in accordance with the experience reported by others.¹⁹ Among 155 consecutive patients followed up for at least 1 year (13.5 ± 9.9 years), there were 15 pseudo-resistant patients in a recent study¹⁹ and we found 16.7% to have pseudo-resistance during a long-term follow-up in a specific epilepsy center.

The presence of all three seizure types of JME was reported to be a risk factor for AED resistance,¹⁹ whereas myoclonus alone²⁰ or a combination of AS and myoclonus are not.²¹ We could not comment on this issue, because our series included two patients having either myoclonus only

or myoclonus and AS and we did not find the presence of all three seizure types quite significant in our small series ($p = 0.059$). It should be noted, however, that a substantial proportion of patients with JME had 5-year remission of myoclonic seizures, although still having rare GTCS.

Another clinical feature associated with AED resistance in JME was the presence of psychiatric problems in our study. A retrospective study of patients with adult-onset idiopathic generalized epilepsy reported that 33% were diagnosed with and treated for mental disorders,²² emphasizing the need for looking for them in patients with IGE. The role of systemic disorders such as thyroid diseases^{23,24} is also important for AED resistance in JME as well as fluctuations in seizure control in an otherwise well-controlled patient as observed in our series.

EEG findings seem not to have any influence on the prognosis. In our first retrospective evaluation with a 7-year follow-up, we found that the normalization of the first diagnostic EEG might have some indication for a good prognosis.⁸ However, our final analysis did not support this initial result.

The first report on patients with intractable JME²⁵ included patients with long duration of epilepsy during which the diagnosis and appropriate treatment was delayed and with a high percentage of asymmetries or focal discharges on EEG. Focal clinical or EEG features or both in patients with JME are common.²⁶⁻³¹ However, we could not confirm that focal changes and long duration or delay of diagnosis have an association with drug resistance in JME.

The most remarkable result of the current study was that the myoclonic seizures in JME had a tendency toward remitting or becoming milder with age. We believe that this tendency is unrelated to the AED treatment or to the lifestyle changes, but we could not rule out possible lasting effects of these interventions. Recently, a clinical study assessed the long-term evolution of progressive myoclonic epilepsy of Unverricht-Lundborg type (EPM1), especially concerning the prognosis of seizures.³² Myoclonia progressed only during the first 5 years; no worsening of myoclonia was seen beyond the 10–15 early years of EPM1.³² Improvement of myoclonia was also observed in some patients with EPM1 in the last 5 years of follow-up and this finding was interpreted by the authors as being probably due to the introduction of new drugs. Even though JME and EPM1 have no similarities regarding pathogen-

esis and course, it is interesting to note that both forms of myoclonic epilepsies have similarity in the alleviation of myoclonus with advancing age.

Recent genetic work suggests that JME may represent several different disorders with different genetic and pathologic signatures found in patients who are clinically indistinguishable or practically so.³³ However, the subtle differences in clinical findings may also be important. It is not easy to know whether patients with remission vs substantial amelioration in myoclonia have a particular genetic signature, or what the relationship may be between the genetic background and prognosis of the individual patient with JME. The roles played by elements such as environmental factors and susceptibility genes remain to be clarified.

We suggest that the unique pathophysiology of the cortical myoclonus of JME improves with time in at least some of the patients though persisting for a long time. There is a marked overlap in the mean ages at remission or alleviation of myoclonia (32.9 ± 9.6 vs 31.3 ± 8.4 years).

Myoclonic seizures are important symptoms, causing impairments and restrictions of social life for some patients with JME at least in the beginning. The patients easily realize that JME is considered as a benign but lifelong genetic disease by the scientific community. We hope that our results showing either a meaningful decrease or remission of myoclonic seizures after a median of 33 years of age will prevent patients with JME and their relatives from losing hope.

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